Dr. Komal Pandya

PhD in Computer Science — Quantum Optimization & Online Algorithms

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Education

Indian Institute of Technology Patna
Ph.D. in Computer Science
Sep 2020 - Sep 2026
Defended: Sep 9, 2026

Thesis: *Higher-Order Matching in 3-Partite, 3-Uniform Hypergraphs: Classical Online Algorithms & Quantum Optimization*

Government Engineering College Gandhinagar
B.E. in Computer Engineering
2015 -2019
Project: *Aspect-Based Sentiment Analysis*

Technical Skills

Quantum Stack
- D-Wave Leap, PennyLane, Qiskit, Cirq

Programming & Tools
- Python, C, MATLAB
- Gurobi, CPLEX, LaTeX

PhD Course Work
- Intro. to Network Science
- Probability, Statistics & Stochastic Processes
- Time Series Analysis
- Random Signals & Systems
- Foundations of Machine Learning
- Artificial Intelligence

Core Computer Science
- Algorithms, Data Structures, Operating Systems, Compiler Design, Automata Theory, Random Matrix Theory

Fellowships & Awards

Quantum Neural Technologies, SA
Research Grant Awardee 2024

Quantum Open Source Foundation Cohort
Research Mentee 2022 – 2024

MHRD PhD Project Fellowship
IIT Patna 2020 – 2023

IndoML 2023, IIT Bombay
Travel Grant Awardee 2023

The CCPWC Projects Final Demonstration, New Delhi
Travel Grant Awardee 2022

Teaching & Mentorship

- **Graduate Teaching Assistant:** Theory of Computation, Foundations of Theoretical Computer Science, Linear Algebra, Computing Lab, Algorithms
- **Research Mentorship:** Co-advised 3+ undergraduate researchers on hypergraph algorithm implementations and quantum simulation pipelines.

Profile Summary

Computer Science PhD specializing in **Quantum Optimization** and **Online Algorithms**, with expertise in QAOA benchmarking, QUBO modeling, hypergraph matching, and multi-objective optimization. Experienced in designing scalable algorithms and analytical frameworks for complex computational architectures. Seeking to tackle challenging problems at the intersection of algorithms, mathematics, and quantum computing.

Research Experience

Quantum Research Mentee Apr 2022 – Mar 2024
Quantum Open Source Foundation (QOSF) *Remote*
- Benchmarked QAOA performance and validated algorithmic scalability across diverse combinatorial problem sets, analyzing the impact of problem size and circuit parameters on solution quality.

IIT Patna PhD Project Fellow Sep 2020 – Aug 2023
MHRD, MHA, Govt. of India *On-Site*
- Developed a web-portal to spread digital awareness for Cyber-Crime Prevention against Women & Children (CCPWC), supporting digital safety and public awareness initiatives.

Quantum Computing Intern Jan 2022 – Apr 2022
Artificial Brain *Remote*
- Formulated and executed a QUBO model for flight trajectory optimization (Deloitte Quantum Climate Challenge 2022), exploring quantum optimization for constrained trajectory planning.

Quantum Computing Intern (QIntern) Jul 2021 – Aug 2021
QWorld *Remote*
- Formulated a QAOA approach for particle tracking in high-energy physics simulations, mapping particle-tracking constraints into a quantum optimization framework.

Machine Learning Intern Jul 2018 – Aug 2018
EISystems *Bengaluru, India*
- Optimized multi-modal ML model performance through systematic hyperparameter tuning and feature extraction, improving predictive performance across multimodal data.

Publications & Manuscripts

- **K. Pandya**, S. Sakhuja, N. Kattaian. "A Topology-Aware Quantum-Classical Framework for Multi-Target Active Debris Removal using Gaussian Boson Sampling." *arXiv preprint arXiv:2610.31824*, 2026.
- **K. Pandya**, A. Maiti. "Benchmarking the Three-Dimensional and the Numerical Three-Dimensional Matching Problems on the D-Wave Advantage Quantum Annealer." *Information Sciences*, 2025.
- K. Blekos, D. Brand, A. Ceschini, CH. Chou, RH. Li, **K. Pandya**, A. Summer. "A review on quantum approximate optimization algorithm and its variants." *Physics Reports*, 2024.
- **K. Pandya**, A. Maiti. Randomized "ϵ-RANKING Algorithm for Online Trichromatic Matching" *IEEE Access*, 2024.
- **K. Pandya**, A. Maiti. "Unweighted Online Matching in k-Regular 3-Partite 3-Uniform Hypergraphs." *Communicated*, 2025.
- **K. Pandya**, A. Maiti. "Edge-Weighted Online Matching in 3-Partite Hypergraphs under Vertex Arrivals." *Communicated*, 2025.
- **K. Pandya**. "Quantum Query Bounds with a Simple Unified Framework for Online Bipartite Matching." *Under Prep*, 2026.

Selected Talks & Presentations

- **Talk:** "Quantum-Integrated Systems for Space Warfare: ISR, C2, and Adaptive Engagement," *SCQC 2025*, Bengaluru.
- **Poster:** "Multi-Target Space Debris Optimization using Gaussian Boson Sampling," *SCQC 2025*, Bengaluru.
- **Talk:** "Advancements in Quantum Warfare," *Conf42*, 2024. (Virtual)
- **Poster:** "Competitive Analysis of Online Trichromatic Matching," *RSD 2023*, IIT Patna.